

Unlock the Potential of High Resolution Spectroscopy on Protoplanets

Insights and Benchmarks from Synthetic Observations

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ABSTRACT

Context. High-resolution spectroscopy combined with high-contrast imaging is a promising route to characterize young giant planets at small separations, where low planet-to-star flux ratios and stellar halo contamination strongly limit detectability.

Aims. We aim to quantify the detectability of young giant planets and key molecular species with VLT/CRIRES+ as a function of planet mass, age, and projected separation, and to provide observational benchmarks for future observations in the ELT era.

Methods. We developed EXOCRIRES, a synthetic observation pipeline that combines atmospheric spectra, ETC-based CRIRES+ simulations, and two post-processing techniques (cross-correlation and likelihood-ratio mapping). We simulated systems at 140 pc with planet masses of 5 and 10 M_{Jup} , projected separations of 24.8–123.9 AU, in multiple evolution stages under a hot-start scenario, then assessed full-spectrum and molecule-specific detectability in the K band.

Results. For 10 M_{Jup} planets, full-spectrum detections above 5σ are achieved at all simulated separations up to ~ 9.5 Myr, with detectability declining toward older/cooler stages and becoming marginal below $T_{\text{eff}} \sim 1400$ K. For 5 M_{Jup} planets, detections at all separations are limited to $\lesssim 3.65$ Myr, with wider separations remaining favorable at later times. H_2O is the most robust tracer, while CO is mainly detectable at high temperatures and CH_4 peaks in intermediate cooling phases. Validation on a β Pictoris b analog reproduces published full-spectrum and H_2O detection levels, while differences in CO/ CH_4 behavior point to the impact of disequilibrium chemistry.

Conclusions. CRIRES+ can robustly characterize very young, massive giant planets in K band with high-contrast spectroscopy, but sensitivity degrades rapidly with cooling and is strongly modulated by orbital separation and chemical assumptions. EXOCRIRES provides a practical framework to define target-selection thresholds and to prepare high-resolution atmospheric studies with current facilities and upcoming ELT instruments.

Key words. giant planet formation – CRIRES+ – synthetic observations

1. Introduction

Observations of young giant planets provide a direct view of planetary systems close to their formation, when retained heat makes these objects substantially brighter than evolved planets at infrared wavelengths (Spiegel & Burrows 2012), and therefore enables direct constraints on quantities relevant to formation scenarios and formation environments. High-contrast imaging (HCI) has rapidly expanded the sample of young planets and strong planet candidates since the first planetary-mass companion detection (Chauvin et al. 2004). Recent facilities, including SPHERE, ERIS, and GRAVITY+, have identified embedded companions such as PDS 70 b, PDS 70 c, WISPIT 2b and WISPIT 2c (Keppler et al. 2018; Haffert et al. 2019; Christiaens et al. 2024; van Capelleveen et al. 2025a). Ongoing surveys and upcoming ELT instrumentation (Chomez et al. 2025; Maio et al. 2025; van Capelleveen et al. 2025b; Liu et al. 2025; Houll   et al. 2021; Feldt et al. 2024) will progressively extend the detection frontier towards planets with closer angular separations ($\lesssim 0.1$ arcsec, i.e. WISPIT 2c, Lawlor et al. 2026),

lower masses and consequently fainter luminosities (i.e., 51 Eri b, Macintosh et al. 2015; Denis et al. 2026).

Characterization of these objects with spectroscopy can help us to precisely constrain their formation scenarios and environment by measuring fundamental atmospheric properties, such as spin rates (Schwarz et al. 2016), thermal structure shown as Pressure-Temperature (P-T) profiles (Aoyama et al. 2021; Berardo et al. 2017), and atmospheric composition including bulk abundances (e.g., C/O ratios) and isotopic ratios (e.g., $^{12}\text{C}/^{13}\text{C}$) (  berg et al. 2011; Zhang et al. 2021; Molli  re et al. 2022). These quantities can be used to distinguish two popular formation scenarios, core accretion (Pollack et al. 1996) and gravitational instability (Boss 1997). Core accretion favors planet formation at inner disks within tens of AU, while the gravitational instability is more efficient in the cold outer disks at larger separations (Kratter & Lodato 2016). Moreover, companions form through a top-down gravitational instability are expected to have similar C/O ratios and bulk metallicity to the host stars due to the fast formation timescales ($\sim 10^3$ years for a Jupiter-mass planet via rapid fragmentation). In contrast, companions formed through a bottom-up core accretion process tend to have a wide range of C/O ratios and metallicities depending on

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the formation location and environment, e.g. inside or outside the CO and H₂O snowlines (Öberg et al. 2011) and the disk chemical composition (Rampinelli et al. 2024).

High Dispersion Spectroscopy (HDS, in this paper we define it as detection speed ≥ 0.1 , with $R \geq 10,000$, from Snellen 2025), in combination with adaptive optics (AO) (dubbed high-contrast spectroscopy in Snellen 2025), is a powerful tool to measure atmospheric properties by dampening the scattering light from the host star to increase the planet-to-star flux ratio, enabling sufficient extraction of the planet signal, and subsequently resolving the molecular lines from planetary atmospheres (Snellen et al. 2015). High-contrast spectroscopy has been applied to several substellar objects including young field brown dwarfs, free-floating planetary-mass companions and super-Jupiters, carried on instruments such as CRIRES+ ($R \sim 100,000$) and KPIC ($R \sim 35,000$), for detailed and population-level atmospheric studies together with atmospheric retrievals. For example, the SupJup survey (de Regt et al. 2024) carried out with CRIRES+ has observed a sample of 38 substellar objects including 17 bound systems, which most of are typically with $M \gtrsim 10 M_{\text{Jup}}$ and located at wide separations > 50 AU (> 0.8 arcsec equivalence in angular separation), as shown in Figure 1 (Zhang et al. 2024; González Picos et al. 2025; Grasser et al. 2025). A systematic atmospheric analysis using KPIC has also been presented in Xuan et al. (2024) for 8 young (1-100 Myr) companions with mass range $10 - 30 M_{\text{Jup}}$ and separation from ~ 50 to 130 AU. Overall, current results of metallicity measurements broadly aligned with the solar value ($C/O \sim 0.59$) and shows marginal variation from field brown dwarfs to Super Jupiter, incapable to constrain formation scenarios as a function of separation. This limitation motivates expanding high-contrast spectroscopy towards lower masses ($\leq 10 M_{\text{Jup}}$) and smaller separations (≤ 50 AU, equivalent to ≤ 0.6 arcsec at 140 pc), where planets may favor a different formation pathway than the wide-separation companions.

High-contrast spectroscopy is promoted to its limit of current instrumentation in this way. First, the luminosity of planets decreases with lower mass at a given age (Marley et al. 2007). Considering the results of a hot-start model in Spiegel & Burrows 2012, K band magnitude of a $5 M_{\text{Jup}}$ planet is ~ 1.5 mag less than a $10 M_{\text{Jup}}$ planet at 1 Myr and ~ 2.8 mag less at 50 Myr. Coupled with rapid cooling, the continuum of spectrum keeps decreasing and the line depths becomes shallower, which further reduces the S/N of planet signal in later ages.

Assuming planets are at certain ages with sufficient luminosity to be captured by current instrumentation, high-contrast spectroscopy is still challenging for two reasons. First, chemical transition during the evolution changes the abundance of molecules at different ages (Burrows et al. 1997; Zahnle & Marley 2014; Moses et al. 2016). This has been observed in the atmosphere of brown dwarfs from L to T types on the shift of primary carbon reservoir, where the CO features gradually weaken while the CH₄ features become more prominent as the temperature decreases (e.g. Testi 2009; Patience et al. 2012). Thus, different molecules favors different ages along the evolution tracks and eventually become undetectable after a certain stage corresponding to a specific effective temperature.

The separation of planets further modulate the detectability. At small angular separations < 1 arcsec, the planetary emission becomes deeply buried in the stellar PSF halo, particularly at separations of a few λ/D even with AO, resulting in a typical flux contrast of $\sim 10^{-2} - 10^{-5}$ in K band, as shown in Figure 2. Together with non-negligible thermal sky background, this yields a typically low $S/N < 0.1$ at the planet position (REFERENCE

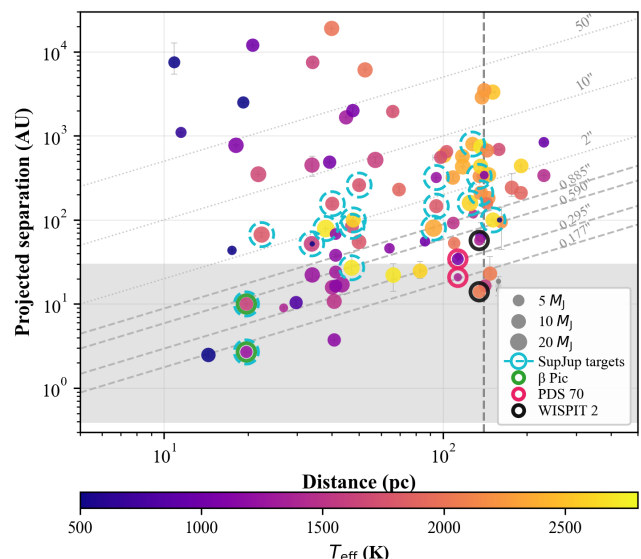


Fig. 1: Confirmed direct-imaging substellar companions in distance–projected-separation space from the NASA Exoplanet Archive (query date: 2026-05-18). Point color encodes planet effective temperature, and point size scales with planet mass. Cyan outlined points mark the cross-matched SupJup targets in 17 bound systems. Three young and proto-planetary systems are highlighted with colored circles: β Pictoris b and c (green), PDS 70 b and c (magenta), and WISPI 2 b (black). The vertical dashed line marks the distance of 140pc. Gray dashed curves mark four selected angular separations, and the shaded band marks the range of semi-major axes in the Solar System.

support). Techniques such as forward modelling and cross-correlation are required to extract the planet signal and retrieve the atmospheric properties (Madhusudhan & Seager 2009; Brogi & Line 2019).

In this paper, we assess the high-contrast feasibility of current instrumentation, specifically CRIRES+, as a function of mass, age and separation of planets, performing a population study of young, direct-imaged giant planets. This requires realistic synthetic observations, and an end-to-end data processing suite. Hence we developed EXOCRIRES, a synthetic observation pipeline that simulates realistic CRIRES+ observations of putative planetary systems with high-contrast spectroscopy. It also adapts the post-processing framework from Landman et al. (2024) for the benchmark target β Pictoris b, as it has a small separations of 0.48 arcsec (Chilcote et al. 2017) and a large planet-to-star contrast of 9 magnitude in K band (Wang et al. 2021), representing a challenging case for CRIRES+ high-contrast spectroscopy at small separations. We then quantify the trend of molecular detectability of H₂O, CO and CH₄ across two evolutionary grids of planetary system with 5 and 10 M_{Jup} planets, separately.

The paper is structured as follows: Section 3 outlines the methodology of the synthetic observation pipeline. Section 4 presents the main results of trends in detectability across the evolutionary grid. Finally, Section 5 discusses our results in the context of current and future instrumentation and outlines limitations and potential extensions of the framework.

Table 1: Parameters and ranges in the evolutionary grid

parameters	Range
Distance d (pc)	140
Projected separation a (AU)	24.8, 41.3, 82.6, 123.9
M_p (M_{Jup})	5, 10
$M_\star$ ($M_\odot$)	1
T_{eff} (K)	(1000, 2400, 100)
[Fe/H]	0.0
$v_p \sin i$ (km/s)	20
$V_\star \sin i$ (km/s)	5
Systematic RV (km/s)	10
Extinction A_V	0
Veiling factor α	0

2. Parameters of Putative Planetary Systems

We first define an evolutionary grid of stellar and planetary parameters for a set of synthetic planetary systems. We fix the distance to 140 pc, typical of nearby star-forming regions such as Taurus (Beckwith et al. 1990), to convert the Ks-band integrated flux into apparent magnitude. The planets are placed at projected separations of 0.177, 0.295, 0.59, and 0.885 arc-sec along the spatial axis, corresponding to physical separations of 24.8, 41.3, 82.6, and 123.9 AU. These values span the range of a sample of direct-imaged planets and candidates are observed to date as illustrated in Figure 1, as well as systems inferred from disk substructures and kinematic signatures (Claudi & Mesa 2025; Long et al. 2019; Izquierdo et al. 2023; Maio et al. 2025; Teague et al. 2025).

We adopt the mass-radius and mass-effective temperature relations for young planets and pre-main-sequence (pre-MS) stars from Baraffe et al. (2015) and Spiegel & Burrows (2012), respectively. Two planet masses, $5 M_{\text{Jup}}$ and $10 M_{\text{Jup}}$ are considered. We focus on the hot-start model from Spiegel & Burrows 2012 because it is the most favorable case for observations. We further assume the stellar age as the system age, and therefore as the planet age.

As illustrated in recent hydrodynamical studies on embedded planets, the extinction of accretion luminosity follows a scaling relation that depends on both the disk surface density and the planet-star separation, and is most likely negligible for a gap-opened planet at Ks band (Alarcón et al. 2024; Sanchis et al. 2020). Here, A_V denotes the line-of-sight extinction caused by dust attenuation, while α represents the veiling fraction of excess emission from disk materials that dilutes spectral lines (González Picos et al. 2025; Alcalá et al. 2021). We focus on gap-opened planet analogues in this work, and hence both A_V and α are set to 0.

We further assume that all simulated systems are face-on, with coplanar planetary orbits, to simplify projection effects, and hence the radial velocity (RV) of the planet and the star are the same. We apply a radial velocity shift of +10 km/s between telluric and planetary lines, corresponding to an Earth's barycentric radial velocity of -10 km/s. We consider 1 $M_\odot$ solar-mass host stars, with stellar metallicity $[\text{Fe}/\text{H}]_\star = 0.0$ (solar), consistent with metallicity measured in the stellar population at Taurus (e.g. Kenyon & Hartmann 1995; Herczeg & Hillenbrand 2014; Biazzo et al. 2011). In addition, we set a high rotational velocity of each planet at 20 km/s following the review of Snellen (2025) on the expected spin rates of young planets as an conservative choice for molecular line detection. The stellar rotational

velocity is set at 5 km/s as a slow rotator, to be consistent with observation values in Taurus (e.g. Nguyen et al. 2012).

The age grid spans planet effective temperature T_{eff} from 2400 K to 1000 K in steps of 200 K. It is chosen to cover the transition from CO-dominated to CH_4 -dominated chemistry (Burrows & Sharp 1999), which can be traced in K band. For 10 M_{Jup} planets, we select eight corresponding ages from 1 to ~64.6 Myr, resulting in K absolute magnitude from ~ x to ~ Y and K-band planet-to-star contrast from ~ 7 to ~ 11, calculated from model spectra described in Section 3.1; for 5 M_{Jup} planets, we select five ages from 1 to 14.9 Myr with K magnitudes decreasing from ~ M to ~ N, resulting in K-band contrast from ~ 9 to ~ 11. Overall, the grid contains 32 simulated systems for 10 M_{Jup} planets and 20 systems for 5 M_{Jup} planets. As plotted in Figure 2, the grids cross a large range of K magnitude and contrast of the current sample, and 10 M_{Jup} planets are considered a popular representative of current direct-imaged sample, while 5 M_{Jup} planets are seen as a lower mass representative that stands at the border line of current detections, which potentially challenges the feasibility of current high-contrast spectroscopy. Table 1 lists the details of the adopted grids. More details of planet and stellar parameters are listed in Appendix C.

3. Methods

We developed EXOCRIRES, a synthetic observation pipeline structured in two modules: (i) generation of mock observations, which generates mock spectra incorporating stellar, planetary, telluric and noise contributions, and (ii) post-processing, which extracts the planetary signal and quantifies detectability. The workflow of this pipeline is shown in Figure 4.

3.1. Atmospheric Model

We use the *Sonora Bobcat* substellar atmospheric models (Marley et al. 2021) to provide the temperature-pressure and volume-mixing-ratio profiles for each putative planet. Although *Sonora* also provides a spectral library, its spectral resolution is insufficient for simulating CRIRES+ observations at $R \sim 100,000$. We therefore feed the temperature-pressure profiles, volume-mixing-ratio profiles, and surface gravity into *PetitRADTRANS-3* (pRT-3, Mollière et al. 2019) to compute high-resolution spectra, which includes three dominant molecular absorbers: CO (HITEMP; Rothman et al. 2010), CH_4 (Hargreaves et al. 2020), and H_2O (POKAZATEL; Polyansky et al. 2018). We also include continuum opacity from H_2 - H_2 and H_2 -He collisional induced absorption (CIA). Host-star spectra are taken from PHOENIX model library assuming $[\text{M}/\text{H}] = 0.0$ (Husser et al. 2013). For both the star and the planet, the gravity is rounded to the nearest 0.5 dex, and the effective temperature is rounded to the nearest 100 K.

As the example illustrated in Figure 3, the relative strengths of molecular bands of CH_4 , H_2O and CO evolve as the planet cools from 1 Myr to 30 Myr for a 10 M_{Jup} planet. The H_2O features remain prominent over the entire temperature range, while CO absorption gradually weakens and CH_4 bands become increasingly pronounced. This trend reflects the chemical transition from a hot, CO-dominated atmosphere to a cooler, methane-rich regime as dictated by equilibrium chemistry for substellar objects (Burrows & Sharp 1999), which is seen from brown dwarfs from L to T types.

In addition, rotational broadening kernels are convolved with the spectra with rotational velocities in Table 1. For stars, the linear limb darkening coefficient ϵ is numerically estimated from

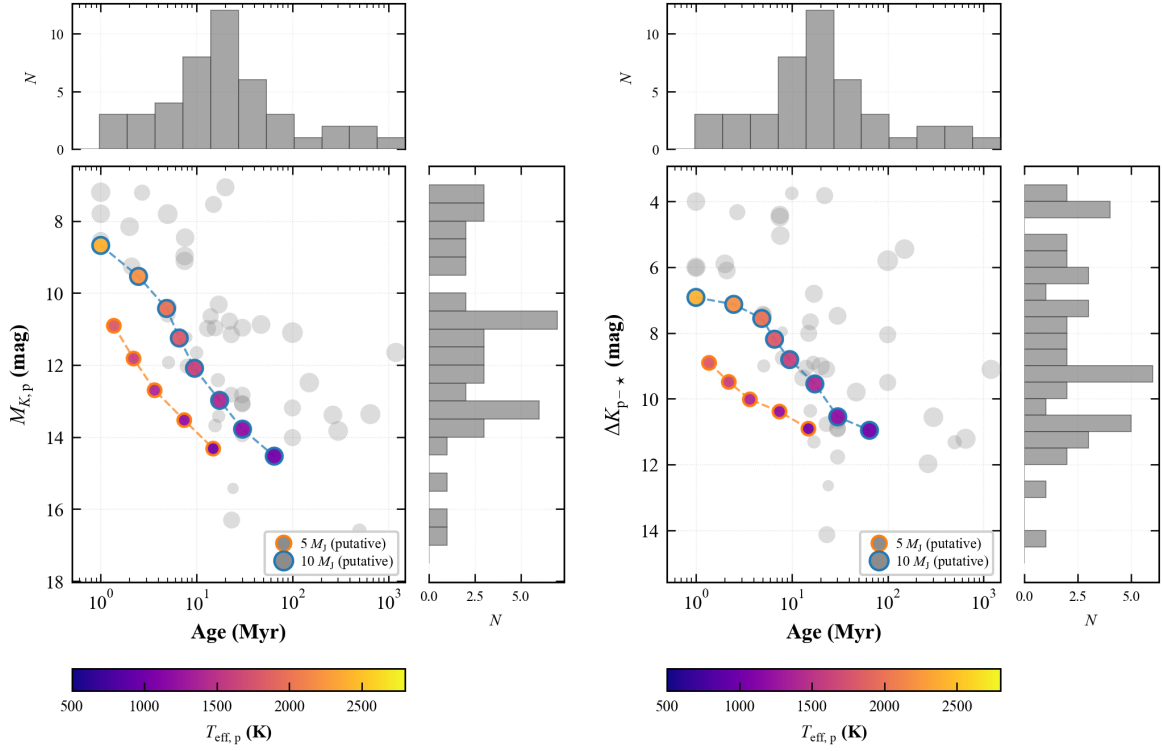


Fig. 2: Evolutionary parameter space of the putative planetary grid compared to the confirmed direct-imaging sample. The left panel shows the absolute K-band planet magnitude $M_{K,p}$ as a function of system age; the right panel shows the K-band planet-to-star contrast $\Delta K = K_p - K_\star$. In both panels, filled circles connected by dashed lines trace the 5 and 10 M_{Jup} evolutionary tracks; marker color encodes planetary effective temperature and marker size scales with planet mass. Gray semi-transparent points show confirmed directly imaged companions with available K-band photometry, also sized by mass. Marginal histograms show the age distribution (top) and magnitude distribution (right) of the confirmed sample.

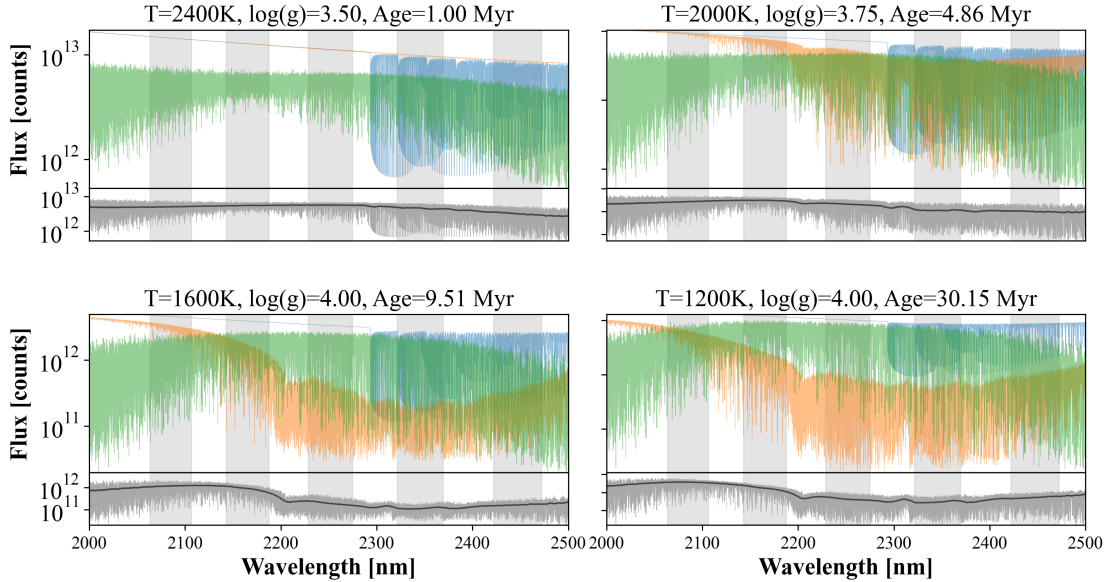


Fig. 3: The colored curves represent CO (blue), CH₄ (orange), and H₂O (green) templates, with the combined spectrum in darkgray in the lower panel of every plot. Grey bands mark the five K2166 spectral orders used in post-processing (orders 1–2 excluded due to strong telluric absorption). The black solid line marks the continuum proxy of the full spectrum using a low-pass filter.

the spectrum using `exotic-1d` (Grant & Wakeford 2024), while the planet ϵ is currently set to zero. The rotational broadened spectra are then calculated by `FastRotateBroad` function from

`PyAstronomy` (Carvalho & Johns-Krull 2023). A Gaussian kernel is also convolved to model the line-spreading function (LSF)

along the dispersion axis, with FWHM equal to average resolution element (λ/R) of CRIRES+.

3.2. Generation of Mock Observation

3.2.1. Simulation Set-up

Our synthetic spectra are used to simulate mock CRIRES+ (Dorn et al. 2023) observations. We oriented the (putative) slit such that the host star was placed at the slit center while also covering the location of the planet. This on-slit configuration results in two distinct spectra along the spatial axis of the detector plane: one from the star and one from the planet, as shown in Figure 5. The stellar spectrum serves as a precise reference for spatial and spectral calibration (e.g. telluric correction and instrument throughput characterization), while the planet spectrum can be extracted differentially with respect to the star.

We simulate atmospheric transmission and telluric lines with parameters related to the observing conditions, including air-mass, turbulence category, fraction of light within slit (FLI), and precipitable water vapor (PWV). The chosen values of these parameters are listed in Table 2, which are selected to represent an optimal weather condition for CRIRES+ high-contrast spectroscopy. Additionally, AO correction was default for all simulations.

EXOCRIRES supports different wavelength settings, and in this paper we focus on K band that covers several spectral features of interest and is less affected by the sky background which stronger in longer wavelengths like the L and M bands. We then applied the K2166 wavelength setting and assumed a slit width of 0.2 arcsec, yielding a spectral resolving power of $R \sim 100,000$. Moreover, the first two of the seven spectral orders were excluded due to the strong telluric absorption, giving an effective wavelength range from 2.06 to 2.47 μm , albeit with gaps between orders.

We also considered the ABBA nodding strategy with two nodding positions. Each nodding cycle consisted of exposure time of one frame of 600s under the saturation threshold of the brightest star in the grid, and then repeated to generate 40 frames with noise distributed for each nodding position (i.e. 40 nodding cycles). This results in a total integration times of 24000s per nodding position, and 48000s (13.3 hours) in total. This choice reflects a balance between maximizing the S/N of the planet signal and maintaining a reasonable observation time for a normal program on VLT.

Then, We fed our model spectra and the parameter values into the CRIRES+ Exposure Time Calculator (ETC) developed by ESO,¹ for the star and planet separately. The ETC outputs a 1-dimensional (1D) integrated spectrum $f(\lambda)$ over the whole extraction aperture width w , modified by instrumental throughputs such as blaze function and instrument efficiency profiles, and the atmospheric transmission profile. **An example ?**

3.2.2. Construction of 2D Spectra

On-slit observations for spatially resolved planetary systems requires two-dimensional (2D) spectra including both signals from the host star and planet, where the horizontal axis represents the wavelength dispersion λ and the vertical axis corresponds to the spatial (or angular) separation x between the star and planet on the sky. We reconstructed synthetic 2D spectra from the 1D ETC outputs and combined both the star and planet signals together.

Table 2: Observing parameters used for the simulations.

Parameter	Value
Exposure time per frame	600s
Nodding cycles	40
Nodding positions	2
Airmass	1.2
Fraction of light within slit (FLI)	0.5
Precipitable water vapor (PWV)	1.5 mm
Turbulence category	70%
Maximum extraction aperture	35 pixels (2.065 arcsec)
Setting	K2166
Reference wavelength	2166.016 nm
Observation mode	NGS AO, Nodding

The 1D output spectrum $f(\lambda)$ can be expressed as the integral of the 2D flux distribution $m(\lambda, x)$ along the spatial axis x :

$$f(\lambda) = \int_{-n}^{+n} m(\lambda, x) dx = s(\lambda)T(\lambda) \int_{-n}^{+n} P(\lambda, x) dx, \quad (1)$$

where $s(\lambda)$ is the spectrum of the target in counts scaled to the input brightness, $T(\lambda)$ is the total transmission including both atmospheric and instrumental throughput; $w = 2n + 1$ is the maximum extraction aperture size in pixels (n), and $P(\lambda, x)$ is the PSF at the spatial position x . To reconstruct the spatial flux distribution, EXOCRIRES varies the extraction aperture size in the ETC output and compute the differential flux between consecutive apertures. Assuming a symmetric PSF centered at $x = 0$ and discrete detector pixels, the flux at a given spatial pixel n is:

$$m(\lambda, \pm n) = \begin{cases} f_1(\lambda), & n = 0 \\ \frac{f_{2n+1}(\lambda) - f_{2n-1}(\lambda)}{2}, & n \geq 1 \end{cases}$$

yielding the 2D flux distribution for both the star and the planet. Therefore, For a single frame, at a given wavelength λ and spatial position x , the flux distribution combined all three photon sources can be expressed as:

$$m_{tot}(\lambda, x) = m_{\star}(\lambda, x) + m_p(\lambda, x) + \mathbf{B}(\lambda) \quad (2)$$

where $m_{\star}(\lambda, x)$ and $m_p(\lambda, x)$ are the flux contributions from the star and planet respectively, and $\mathbf{B}(\lambda)$ is the sky background flux which we assume to be uniform across the slit and hence only depends on wavelength. Notably, the planet flux $m_p(\lambda, x)$ is spatially offset from the star by a certain number of pixels corresponding to the projected separation x_p , as shown in the right panel of Figure 5.

3.2.3. Noise Sources

Noise in each detector pixel arises from two main sources: Poisson noise from photon counts (stellar, planetary, and sky background) plus dark current, and Gaussian read-out noise (RON). For a pixel at position (x, λ) in a single-frame 2D spectrum, the total signal distribution $F(x, \lambda)$ is:

$$F(x, \lambda) = \mathcal{P}[m_{tot}(\lambda, x) + \mathbf{D}] + \mathcal{N}(0, \sigma_{\text{RON}}^2) \quad (3)$$

where $\mathcal{P}$ denotes Poisson distribution; $\mathbf{D}$ is the dark current per pixel, and $\mathcal{N}$ represents Gaussian noise with standard deviation σ_{RON} . All noise parameters are obtained from the CRIRES+ ETC, while $\mathbf{D} = 1.8 e^-/\text{pix}$, $\sigma_{\text{RON}} = 6 e^-/\text{pix}$ for a 600-second exposure.

¹ <https://etc.eso.org/observing/etc/crises2>

The final 2D mock spectra are cumulatively stacked over all nodding pairs following the ABBA subtraction to remove background emission. An example is shown in the left panel of Figure 5.

In our simulations, dark current and read-out noise are negligible compared to the photon noise. Thus, after stacking all nodding frames and subtracting stellar light, the signal-to-noise ratio at the planet position can be approximated as:

$$S/N \sim \frac{\sqrt{N_{\text{DIT}}} \cdot m_p(\lambda, x_p)}{\sqrt{m_p(\lambda, x_p) + m_\star(\lambda, x_p) + \mathbf{B}(\lambda) + \eta(\lambda, x_p)}} \quad (4)$$

where $\eta(\lambda, x_p)$ represents residual flux after stellar subtraction.

3.3. Data Analysis

The principle of analysis for high-contrast spectroscopy is based on linearity, where at each spatial position of the observed spectrum, the signal can be expressed as a linear combination of the stellar and planetary contributions, plus noise and potential systematics. And therefore, we aim to simultaneously disentangle the planet and stellar contributions from 2D datasets, and then apply likelihood-ratio mapping on the extracted planet spectrum to quantify the detection. EXOCRILES adapts a state-of-the-art data modeling and extraction approach is proposed by Ruffio et al. (2019), and recently applied on direct-imaged targets e.g. Landman et al. (2024) and González Picos et al. (2025).

As we focus on mock observations without instrumental systematics, we do not incorporate the modeling of systematics in the current framework. We also assume no prior knowledge of the planetary and stellar spectra to mimic the real observation scenario. In other words, the data analysis module is blind to the input model spectra, and the stellar and planetary contributions are estimated solely based on the observed data. Then, the flux in the 2D spectra can be considered as comprising two primary components: emission from the planet and from its host star. The spectrum at a specific spatial position, denoted as $\mathbf{d}$, can thus be expressed as:

$$\mathbf{d} = M_\psi \mathbf{c} + \eta = [\mathbf{m}_p(\psi), \mathbf{m}_s(\psi)] \times [c_s, c_p] + \eta \quad (5)$$

Here, M_ψ represents the model matrix including both component, and η is the noise term. The stellar contribution $\mathbf{m}_s$ is specifically constructed as

$$\mathbf{m}_s = \frac{\mathcal{L}[\mathbf{d}]}{\mathcal{L}[\mathbf{f}_s]} \mathbf{f}_s \quad (6)$$

where $\mathcal{L}$ refers to a low-pass filter to extract the low-frequency features from the spectrum, $\mathbf{f}_s$ is the 1D master template, obtained by integrating over the central three pixels at the stellar position to minimize contamination from planetary flux leakage. Hence, the ratio $\frac{\mathcal{L}[\mathbf{d}]}{\mathcal{L}[\mathbf{f}_s]}$ acts as a scaling factor for the master template to match with the mock stellar spectrum at a given spatial position. Notably, tellurics lines and transmission profiles are not separately corrected or modeled in the stellar contribution but are inherently included within $\mathbf{m}_s$.

Then, for the planet contribution $\mathbf{m}_p$ we can write:

$$\mathbf{m}_p(\psi) = T(\lambda) \mathbf{F}_p(\lambda, \psi) - \frac{\mathcal{L}[T(\lambda) \mathbf{F}_p(\lambda, \psi)]}{\mathcal{L}[\mathbf{f}_s]} \mathbf{f}_s \quad (7)$$

where the total transmission profile $T(\lambda)$ is extracted from the 1D master template divided by a stellar model spectrum from PHOENIX, and here we use the input stellar model; $\mathbf{F}_p$ is the

planet model spectrum. Therefore, the second term in Equation 7 functions as a correction of the planet continuum, ensuring that the modeled planetary signal aligns accurately with the observed data.

In Equation 5, $\mathbf{c} = [c_s, c_p]$ are linear scaling coefficients and can be estimated by comparing the model and the data. In the subsequent analysis we solve for $\mathbf{c}$ and map the detection significance with the likelihood-ratio framework described below, aiming to optimize the detection and quantification of the key molecules in the planet atmosphere. A cross-correlation analysis is additionally carried out as an independent validation of the likelihood results (Appendix A).

In the likelihood-ratio mapping workflow, we apply the linear least square solution $\hat{\mathbf{c}}$ to estimate the scaling coefficients at each spatial position and velocity shift. The solution is given by:

$$\hat{\mathbf{c}} = (M_\psi^T \Sigma_0^{-1} M_\psi)^{-1} M_\psi^T \Sigma_0^{-1} \mathbf{d} \quad (8)$$

And then the χ^2 of the fit is determined as:

$$\chi_0^2 = (\mathbf{d} - M_\psi \hat{\mathbf{c}})^T \Sigma_0^{-1} (\mathbf{d} - M_\psi \hat{\mathbf{c}}) \quad (9)$$

Then, the likelihood of the model parameters ψ given the data $\mathbf{d}$ can be calculated based on the Gaussian likelihood function. Eventually, we adopt the final modified posterior function proposed by Ruffio et al. (2019) and Landman et al. (2024) which accounts for a prior on the noise scaling parameter. Following Equation 15 from Landman et al. (2024):

$$\mathcal{L}(\psi | \mathbf{d}) = \frac{P(\psi)}{\sqrt{|\Sigma_0|} \times |M_\psi^T \Sigma_0^{-1} M_\psi|} \left(\frac{1}{\chi_0^2} \right)^{\frac{N_d - N_c + \gamma - 1}{2}}, \quad (10)$$

$P(\psi)$ represents the prior on the model parameters, and is fixed to 1 as a non-informative uniform prior in this work. Σ_0 is the data covariance matrix, and M_ψ describes the forward model matrix. χ^2 quantifies the goodness of fit. N_d and N_c represent the number of data points and the number of components respectively. In Ruffio et al. (2019) they introduced a improper prior for noise scaling parameter s , which is $P(s) \propto s^{-\gamma}$ while $\gamma = 2$ assuming Gaussian noise in Landman et al. (2024). Hence, since Σ_0 doesn't depend on ψ , numerically the log posterior can be expressed as:

$$\begin{aligned} \ln \mathcal{L}(\psi | \mathbf{d}) &= -\frac{1}{2} \ln |\Sigma_0| - \frac{1}{2} \ln |M_\psi^T \Sigma_0^{-1} M_\psi| - \frac{N_d - N_c + 1}{2} \ln \chi_0^2 \\ &\propto -\frac{1}{2} \ln |M_\psi^T \Sigma_0^{-1} M_\psi| - \frac{N_d - N_c + 1}{2} \ln \chi_0^2 \end{aligned} \quad (11)$$

at each velocity shift. For details of derivations, see Appendix D in Ruffio et al. (2019).

The detection significance is assessed using a log-likelihood ratio test. Under the null hypothesis H_0 , the planet amplitude is fixed at $c_p = 0$, retaining only the stellar contribution. For each trial velocity v_i , we compute:

$$\Delta \ln \mathcal{L}(v_i) = \ln \mathcal{L}_{v_i, \text{nominal}} - \ln \mathcal{L}_{v_i, \text{null}} \quad (12)$$

where the nominal model includes both stellar and planetary contributions, and the null model contains only the stellar component. A larger $\Delta \ln \mathcal{L}$ indicates stronger evidence for a planetary signal. To isolate individual molecular contributions, the likelihood ratio is adapted to:

$$\Delta \ln \mathcal{L}_X(v_i) = \ln \mathcal{L}_{v_i, \text{nominal}} - \ln \mathcal{L}_{v_i, \text{without } X} \quad (13)$$

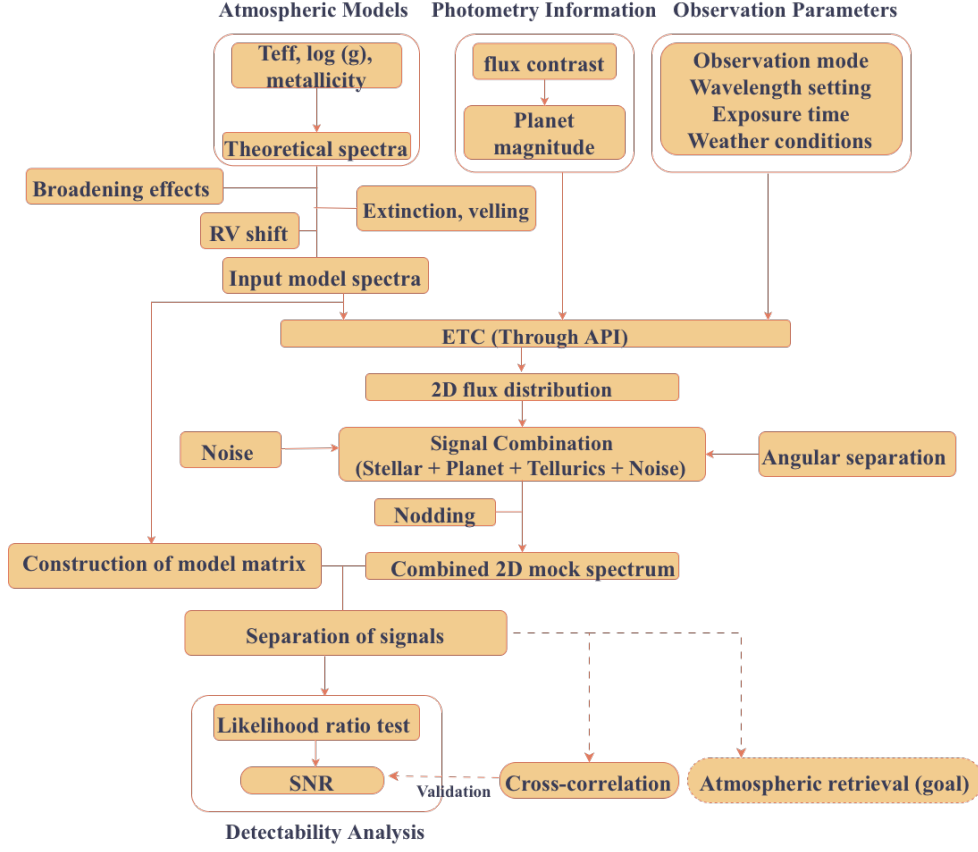


Fig. 4: Overview of the EXOCRILES pipeline. Inputs include atmospheric model spectra, planet magnitudes measured via HCI flux contrast, and CRIRES+ ETC parameters (disk-related effects such as extinction and veiling are set to zero in the present work). Outputs feed into the likelihood-ratio mapping framework used for the detectability analysis, with cross-correlation retained as an independent validation step (Appendix A). The dashed outline indicates the atmospheric retrieval module as a goal procedure in future work and not implemented in this work.

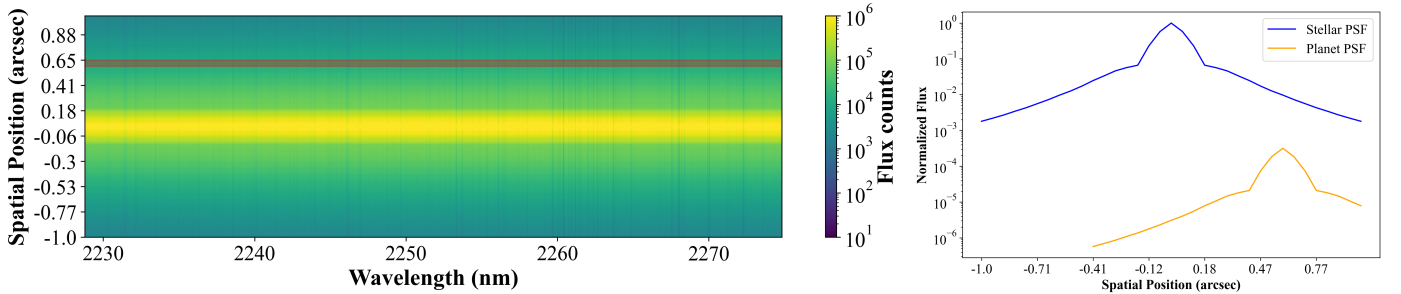


Fig. 5: A mock 2D spectrum generated from the pipeline for a $10 M_{\text{Jup}}$ putative planet, with $T_{\text{eff,p}} = 1800\text{K}$ located at 82.6 AU, marked with red dashed line on the left panel. The planet emission is strongly covered by the stellar PSF halo and thus not visible here. On the right panel it shows the model PSF profiles for the star (blue) and planet (orange) at spectral Order-25 (2283.474 - 2325.300 nm), normalized to the peak of the stellar PSF.

where X denotes one of the three molecular species in the atmospheric model, and the model without X serves as the alternative null. Both quantities are evaluated over $v_i \in [-100, 100]$ km/s at each spatial position along the slit, as illustrated in Figure 6.

At the planet position, $\Delta \ln \mathcal{L}(v_i)$ peaks at the injected radial velocity and is elevated across the full velocity range. The low-frequency continuum of the planet spectrum is partially absorbed into the stellar scaling coefficient c_s at off-peak velocity shifts,

yielding a non-zero $\Delta \ln \mathcal{L}$ far from the true velocity. We remove this bias by subtracting the peak value at injected velocity of 10 km/s as a normalization:

$$\Delta \ln \mathcal{L}_{\text{norm}}(v_i) = \Delta \ln \mathcal{L}(v_i) - \Delta \ln \mathcal{L}_{\text{max}} \quad (14)$$

Under Wilks' theorem (Wilks 1938), the statistic $2 \Delta \ln \mathcal{L}_{\text{norm}}$ follows a χ^2 distribution with one degree of freedom, since c_p is the sole additional free parameter in the nominal model rela-

tive to the null. The Gaussian detection significance is then:

$$z(v_i) = \sqrt{2 \Delta \ln \mathcal{L}_{\text{norm}}(v_i)} \quad (15)$$

The significance is evaluated across the evolutionary grid in Section 4 to establish detectability trends as a function of planet mass, separation, and age. A detection is claimed when $z(v_i)$ exceeds 5σ at $RV = 10$ km/s.

In this evolutionary study with a grid of planetary systems, we adopt the likelihood-converted significance z as the primary detectability metric. Although Σ_0 appears explicitly in the likelihood, its absolute scale does not affect the detection significance. Since the same Σ_0 enters both the nominal and null likelihoods, the $\frac{1}{2} \ln |\Sigma_0|$ terms cancel identically in $\Delta \ln \mathcal{L}$. For the molecular detection ratio $\Delta \ln \mathcal{L}_X$, where both the nominal and alternative null models carry identical numbers of components $N_c = 2$, any uniform multiplicative error in Σ_0 further cancels from both the χ^2_0 and determinant contributions, leaving the ratio exactly scale-invariant. For the full-spectrum ratio, where the nominal model has $N_c = 2$ and null models have one component with $N_c = 1$, a uniform scale error in Σ_0 introduces a velocity-independent additive constant to $\Delta \ln \mathcal{L}$; this constant is subsequently removed by the peak normalization in Equation 14, so the final significance z is equally unaffected. The detection significance is therefore robust against systematic errors in the absolute calibration of Σ_0 .

4. Results

In this evolutionary study, we identify two key factors that govern the trend of detectability for a given mass of planet: (i) The age of the system, which determines the intrinsic luminosity of both planet and host star, and hence the planet-to-star contrast. The age also sets the planet's effective temperature, which determines the presence and strength of molecular absorption features in the planet spectrum given a certain chemistry regime. (ii) The configuration of the system, which reflects the spatial separation between the star and planet on the detector, further modulates the order of scattering starlight in the planet signal and hence the S/N of planet signal. The trend of detectability across the whole grid is thus a interplay between these two factors, unveiled by Figure 7 and Figure 8.

4.1. Full-spectrum Detectability

Figure 7 shows the full-spectrum detectability significance (σ) as a function of age for $10 M_{\text{Jup}}$ (right panel) and $5 M_{\text{Jup}}$ (left panel) planets at different orbital separations. For both masses, the significance decreases monotonically with age, reflecting the expected decline in luminosity as the planets cool and fade. In the $10 M_{\text{Jup}}$ case, young planets are detectable at all separations (24.8-123.9 AU) for ages $\lesssim 9.5$ Myr, corresponding to K magnitude $\lesssim 17.8$ mag and $T_{\text{eff,p}} \lesssim 1600$ K. As the planets cool below ~ 1400 K with ages $\gtrsim 17.4$ Myr, the detectability progressively approaches the 5σ threshold from smaller to larger separations. No planet remains detectable at ages ≥ 64 Myr, when $K = 20.1$ mag and $T_{\text{eff,p}} = 1000$ K.

A similar trend is found for $5 M_{\text{Jup}}$ planets that become detectable at younger ages. Detectability at all selected separations is limited to ages $\lesssim 3.7$ Myr. Different from the $10 M_{\text{Jup}}$ case, $5 M_{\text{Jup}}$ planets at the largest separation (123.9 AU) remain detectable down to the lowest effective temperatures, with the 5σ detection limit occurring below $T_{\text{eff,p}} = 1000$ K.

4.2. Observability of Molecules

Figure 8 presents the molecular observability significance (σ) for CH_4 , CO , and H_2O as a function of age and separation. The equilibrium chemistry model has probed that different species favor different temperature ranges of the thermochemical equilibrium sequence: CO is favored in the hottest atmospheres, CH_4 emerges only after sufficient cooling, and H_2O remains the dominant gas-phase oxygen reservoir across a wide range of surface gravities and temperatures down to the water condensation temperature (~ 300 K) (Burrows & Sharp 1999; Lodders & Fegley 2002). Moreover, according to the thermal evolution tracks, the absorption line depths decline consequently with age due to the continuous cooling. Coupled with both effects, the observability trends in Figure 8 provide a direct diagnostic of CRIRES+ feasibility across different ages for key molecules in planetary atmospheres.

H_2O is the most robust tracer of atmosphere in both 5 and $10 M_{\text{Jup}}$ cases. The signals are above 5σ threshold over a much broader range of ages than those for CO and CH_4 . For $10 M_{\text{Jup}}$ planets, H_2O remains above the 5σ threshold at all separations through 17.4 Myr, corresponding to $T_{\text{eff}} = 1400$ K. For $5 M_{\text{Jup}}$ planets, it is detectable at all separations through 3.7 Myr and remains detectable at the widest separation (123.9 AU) down to 14.9 Myr, when $T_{\text{eff}} = 1000$ K with K magnitude $\gtrsim 20$ mag.

CO is much more restrictive. For the $5 M_{\text{Jup}}$ grid, the initial temperatures are already too low for CO to be detected, where the signal is only marginally above 5σ at ~ 1 Myr, the youngest age, and at the largest separation. For $10 M_{\text{Jup}}$ planets, CO can be traced at all separations up to 4.9 Myr, when $T_{\text{eff}} = 2000$ K. Beyond this age the signal weakens quickly: it is limited to the largest separation at ~ 6.8 Myr and eventually falls below 5σ at all selected separations by 9.5 Myr, when $T_{\text{eff}} = 1400$ K and K magnitude ~ 18 mag.

For CH_4 , a certain observation window is seen for $10 M_{\text{Jup}}$ planets from 2.2 Myr to 9.5 Myr, with T_{eff} from 2200 K down to 1600 K, while CH_4 is not detectable at the youngest age when the planet is hottest, as the equilibrium chemistry favors CO over CH_4 ; CH_4 signals then appears with high significance at 2.2 Myr with $T_{\text{eff}} = 2200$ K, with $\sigma > 10$ in all separations. The signal then weakens with age but remains detectable until 9.5 Myr, with T_{eff} down to 1600 K. After 9.5 Myr, although the temperature favors formation of CH_4 , the planet is too faint (K magnitude $\gtrsim 18$ mag and $T_{\text{eff}} < 1600$ K), and the decreasing continuum gradually reduces the line depths below the 5σ threshold. At large separations of 82.6 and 123.9 AU, CH_4 signal keeps a significance of $\sim 4 - 6\sigma$ from 17.4 Myr to 30.2 Myr, and no longer detectable in later ages. For $5 M_{\text{Jup}}$ planets, CH_4 is already marginal at young ages < 5 Myr, and only remains weakly detectable at the widest separation in older ages of the grid. Overall, in an atmosphere following the equilibrium chemistry, the most useful age range for constraining C/O ratios when all three molecules are simultaneously detectable is therefore between ~ 2 to 10 Myr for $10 M_{\text{Jup}}$, and younger than 5 Myr for $5 M_{\text{Jup}}$ planets, approximately.

5. Discussion

5.1. Detectability across the population of young, self-luminous planets

The trend of detectability in Figure 7 and 8 can be scaled and applied to the population of young, self-luminous planets. The current direct-imaging sample within the largest separation in

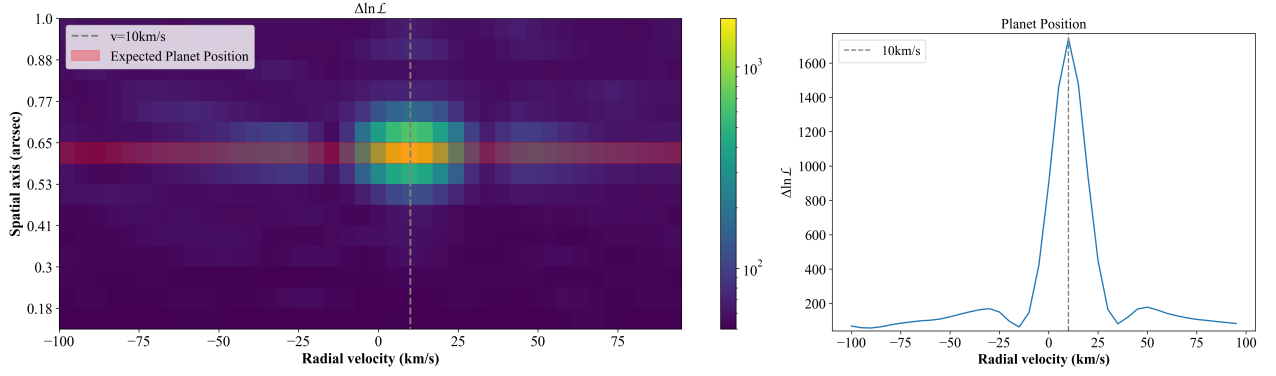


Fig. 6: *Left*: Log-likelihood ratio ($\Delta \ln \mathcal{L}$) map as a function of trial velocity and spatial position along the slit. The red dashed line marks the expected planet position along the spatial axis; the gray dashed line marks the injected radial velocity of 10 km/s. *Right*: Corresponding 1D $\Delta \ln \mathcal{L}$ profile extracted at the planet position, showing a significant peak at the injected velocity.

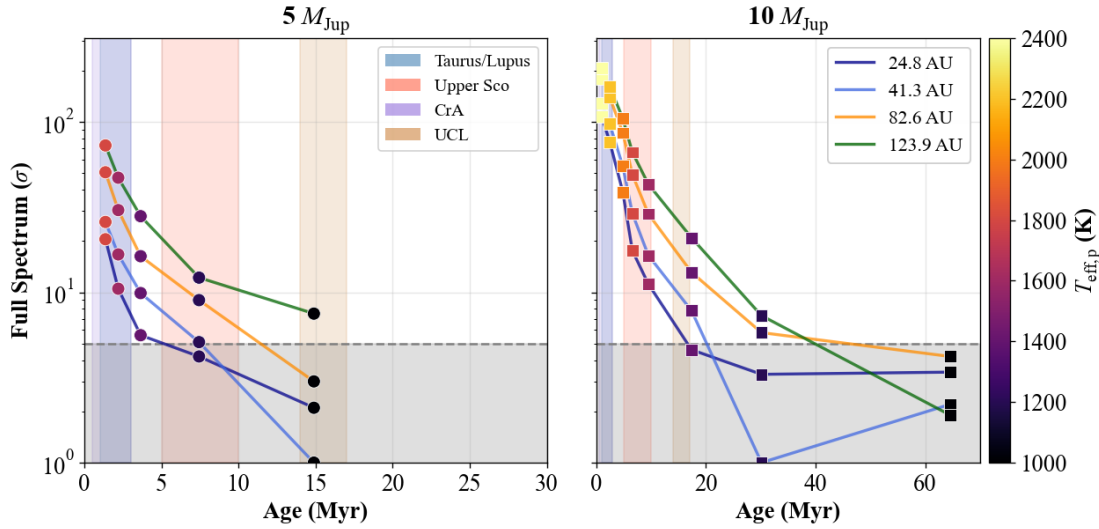


Fig. 7: Full-spectrum detectability significance (σ) as a function of age for planets with masses of $10 M_{\text{Jup}}$ (right panel) and $5 M_{\text{Jup}}$ (left panel). Each curve corresponds to a different orbital separation, and the color scale of data points indicates the planet's corresponding K magnitude (effective temperature) according to Appendix C. The horizontal dashed line marks the 5σ detection threshold.

our grid (0.885 arcsec) already spans a wide range of masses, ages, and temperatures. The detectability curves in Figure 7 and Figure 8 can therefore be mapped directly to a fraction of known systems including PDS 70 b and c, AB Aur b, and WISPIT 2 b and c. In particular, WISPIT 2 c is challenging for CRIRES+ K-band feasibility due to its small separation (~ 0.1 arcsec, 2 pixels on the detector) but still within the reach. Focusing on the 4.9 Myr-old, $10 M_{\text{Jup}}$ putative planet located at 24.8 AU in our grid, with $\Delta K \sim 7.6$ that shares similar parameters from measurement of WISPIT 2 c, we can approximately scale the detection significance by reducing $\sim 24\%$ due to the increased stellar contamination moving from 3 to 2 pixels, according to the PSF profile in Figure 5 and scaling relation in Table 3. In this way, CH_4 is still expected to be detected at $\sim 7.6\sigma$, while the H_2O remains a strong detection with significance at $\sim 26\sigma$. CO is weak and has a marginal detection at $\sim 5\sigma$ level. Nevertheless, this optimistic estimation is again based on the idealized scenario of a gap-opened environment with negligible extinction and veiling from protoplanetary and circumplanetary disks, and given an equilibrium chemistry regime of the planetary atmosphere.

5.2. Inspirations for Future Surveys and Target Selection

The nearby star-forming regions at ~ 140 pc marked in Figures 7 and 8, Taurus/Lupus ($\sim 1\text{--}3$ Myr), CrA ($\sim 0.5\text{--}3$ Myr), Upper Sco ($\sim 5\text{--}10$ Myr), and UCL ($\sim 14\text{--}17$ Myr), overlaps with the detectable range in our evolutionary grid (). Giant planets imaged in these regions are prime targets to be followed up by CRIRES+ for characterization; the coverage extends to UCL at $14\text{--}17$ Myr, where H_2O features in $10 M_{\text{Jup}}$ planets' atmospheres remain detectable with a significance $> 5\sigma$ at separations > 24.8 AU/0.177 arcsec, motivating target selection across a broader age baseline in future high-contrast spectroscopy surveys.

Upcoming ELT instruments will further expand the directly imaged planet sample within the age and separation range of our grid. HARMONI's high-contrast module provides K-band integral-field spectroscopy at $R = 17,000$ reaching $\Delta K \sim 15$ mag at separations of $50\text{--}400$ mas and gaining ~ 2.5 mag over classical HCI through molecule mapping (Houll   et al. 2021). METIS adds L- and M-band high-contrast imaging and $R \approx 100,000$ IFU spectroscopy (Feldt et al. 2024), extending sensitivity to cooler companions. New detections from both instruments within our

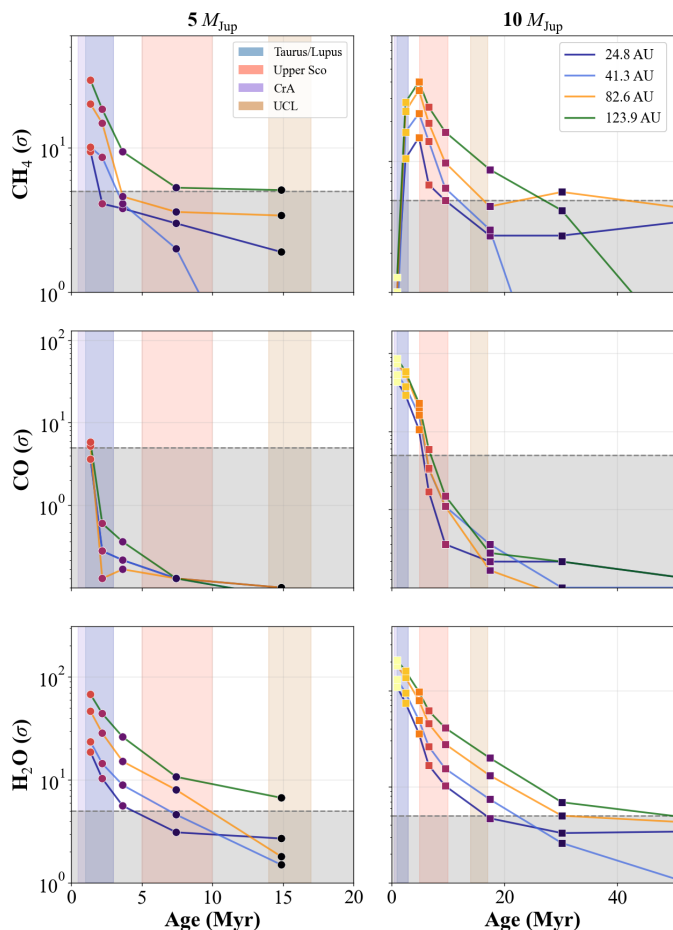


Fig. 8: Same layout as Figure 7, but showing the detectability significance of three key atmospheric molecules included in the modelling: CH₄ (top row), CO (middle row), and H₂O (bottom row), for 5 M_{Jup} (left column) and 10 M_{Jup} (right column) planets at different orbital separations. The dashed horizontal line marks the 5 σ detection threshold.

grid boundaries can be promising candidates for CRIRIS+ characterization in K band.

For companions at separations closer, or older/fainter than the detection limits in our grid, two complementary strategies are actively being demonstrated. ERIS/SPIFFIER delivers K-band integral-field high-contrast spectroscopy at $R \approx 11\,000$, enabling molecular mapping down to ~ 0.12 arcsec and $\Delta K \sim 12$ mag. Hayoz et al. 2025 applied this to AF Lep b ($3.75 \pm 0.5 M_{\text{Jup}}$, $\Delta K = 11.84$ at 0.32 arcsec), detecting H₂O and CO at 9.3σ and 9.7σ . For precise spin measurements and radial velocity constraints, HiRISE couples SPHERE to CRIRIS+ via a single-mode fiber, delivering H-band spectroscopy at $R \approx 140,000$ at H band. Denis et al. 2026 applied this to 51 Eri b ($\lesssim 9 M_{\text{Jup}}$, Age $\sim 24 \pm 3$ Myr, ~ 0.2 arcsec) and constrained a highly eccentric orbit ($e = 0.55^{+0.03}_{-0.07}$) from high-precision radial velocity measurements. These approaches extend characterization of planet properties to a regime beyond the separation ranges explored in this work focusing on CRIRIS+.

EXOCIRIS is not restricted to CRIRIS+ or the K band; given an ETC and a PSF model, it can be adapted to any slit, fiber or integral-field spectrograph to compare detectability thresholds across instruments and support simulations for ELT-era instruments on high-contrast spectroscopy.

5.3. Noise contributions

The noise at the position of the planet is a combination of photon noise from the star, planet and sky (Birkby 2018). The relative contribution of these noise sources is different across the evolutionary grid and separations. The relative contribution of stellar photon noise over total noise can be estimated by:

$$\frac{\sqrt{F_{\star,x}}}{\sqrt{F_{\star,x} + F_{p,\text{peak}} + \mathbf{B}}} \quad (16)$$

and in stellar-limit regime it is approximately 1. Figure 9 shows the stellar noise fraction at the four selected separations. At small separations $\lesssim 0.12$ arcsec, stellar photon noise dominates across all ages. Moving outwards, sky background gradually becomes a non-negligible contributor for systems older than ~ 2.5 Myr, with the stellar noise fraction falling below 90% at ~ 0.42 arcsec, and reaching $\sim 50\%$ for the oldest systems (~ 64.58 Myr) at 1.0 arcsec. The noise budget at large separations is therefore more complex than a purely stellar photon-limited regime, indicating limited gain from AO suppression of starlight and the importance of accurate background subtraction for faint host stars at wide separations. In addition, For young planets at age $\lesssim 4.9$ Myr with $T_{\text{eff}} \gtrsim 2000$ K, the planet photon noise has a notable contribution at large separations $\gtrsim 0.59$ arcsec. At ages older than ~ 6.6 Myr with $T_{\text{eff}} < 1800$ K, the planet flux decreases and its noise contribution becomes negligible at all separations. We summarize the S/N scaling relation in different noise regimes in Table 3, which can be used to approximately scale the detectability significance for planets with different fluxes and separations from the evolutionary grid.

5.4. Idealized Assumptions in the Evolutionary Grid

5.4.1. Extinction and veiling

In this evolutionary study, we focus exclusively on planetary systems with negligible extinction and veiling, thereby representing optimistic observational conditions. The adopted configuration consists of a single, post-formation, gap-opened giant planet orbiting a solar-mass pre-main-sequence star at the distance of 140 pc with solar metallicity, and assuming a hot-start formation scenario. This setup constitutes an idealized case compared to currently detected planets embedded in protoplanetary disks which shows non-negligible extinction effects i.e. PDS 70 b (Wang et al. 2021).

The additional extinction from both circumstellar and circumplanetary disk material is set to zero at zeroth order. For gap-opening planets with $M_p/M_\star \gtrsim 10^{-3}$ at large orbital radii, K-band extinction is $A_\lambda < 0.2$ mag and becomes negligible at inclinations $\gtrsim 60^\circ$ (Sanchis et al. 2020; Alarcón et al. 2024), supporting our assumption of $A_V = 0$. This assumption may break down at high disk inclinations or in denser, less-evolved disk environments, where dust column density and snow-line chemistry can raise the local opacity toward the planet, and incorporating realistic A_λ profiles across disk evolution stages simultaneously with star and planet evolution, remains a natural extension of this work.

Although emission from the circumplanetary disk (CPD) is not expected to dominate in the K band from apparent SED fitting studies on brown dwarfs and giant planets (Stolker et al. 2021; Mâlin et al. 2025; Christiaens et al. 2024), ongoing accretion may nevertheless introduce significant continuum veiling, analogous to that observed in T Tauri stars (Hartigan et al.

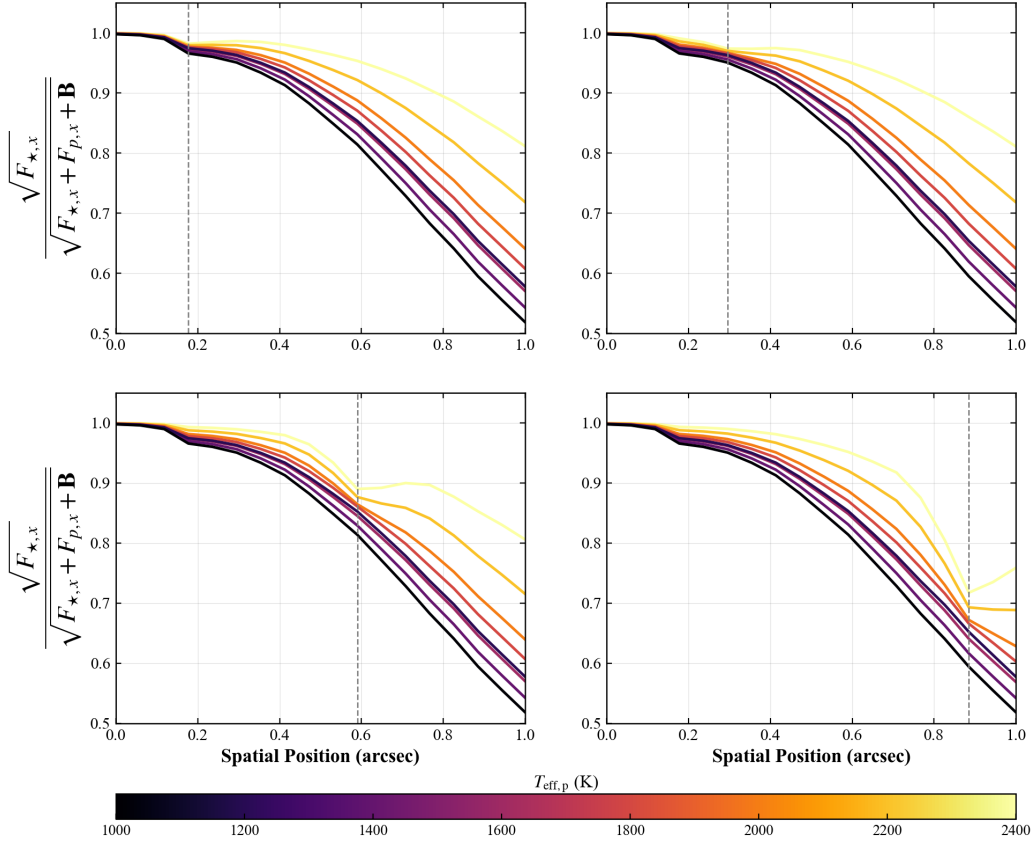


Fig. 9: Stellar photon-noise fraction $\sqrt{F_{\star,x}} / \sqrt{F_{\star,x} + F_{p,x} + \mathbf{B}}$ as a function of angular separation. Each curve traces one planet in the evolutionary grid, coloured by effective temperature $T_{\text{eff},p}$. The grey dashed vertical line marks the injected planet position in each panel.

Table 3: Dominant noise regime and approximate S/N scaling, adapted from Equation 4 at the planet position as a function of angular separation and planet effective temperature (age), for $10 M_{\text{Jup}}$ planets in the evolutionary grid (Figure 9).

Separation (arcsec / AU at 140 pc)	Age ($10 M_{\text{Jup}}$) / $T_{\text{eff},p}$	S/N
$\lesssim 0.12 / \lesssim 16.8$	all / all	$\propto \sqrt{F_{\star,x}}$
$0.12\text{--}0.59 / 16.8\text{--}82.6$	all / all	$\propto \sqrt{F_{\star,x} + \mathbf{B}}$
$\gtrsim 0.59 / \gtrsim 82.6$	$\gtrsim 6.6 \text{ Myr} / \lesssim 1800 \text{ K}$	$\propto \sqrt{F_{\star,x} + \mathbf{B}}$
$\gtrsim 0.59 / \gtrsim 82.6$	$\lesssim 6.6 \text{ Myr} / \gtrsim 1800 \text{ K}$	$\propto \sqrt{F_{\star,x} + \mathbf{B} + F_{p,\text{peak}}}$

1989). Strong veiling reduces the depth of photospheric absorption lines, thereby directly decreasing sensitivity to key molecular tracers. In the present work, the planets are treated as post-formation objects without substantial ongoing accretion, and the veiling contribution is therefore assumed to be negligible. A more comprehensive assessment would ideally explore a parameterized range of veiling factors at very early ages, i.e. $\sim 1\text{--}2$ Myr grids across the formation timescales of giant planets.

5.4.2. Atmospheric chemistry

We include three essential species CO , H_2O , and CH_4 in our atmospheric modelling. The relative abundance of these species is determined by the thermal structure of the atmosphere, which is in turn set by the planet's effective temperature and surface gravity. Real observations of hottest direct-imaged planets have revealed other species such as HF and atomic species including Ca and Na (González Picos et al. 2024). One could run EX-

OCRILES with an atmosphere model including more species to explore the detectability across the evolutionary grid.

Furthermore, we assume chemical equilibrium without quenching of CH_4 and cloud formation in modelling. As K band favors the absorption at high altitudes of the atmosphere, the presence of cloud opacity may not strongly affect the line depths and the consequent detectability of molecules (Snellen 2025). Quenching effects can significantly alter the abundance of CH_4 and CO . For instance, β Pictoris b has an effective temperature of $\sim 1700 \text{ K}$, $\Delta K \sim 9.1$ and angular separation ~ 0.48 arcsec in 2021, where the equilibrium chemistry favors CH_4 detection according to Fig 8. However, Landman et al. 2024 reported a non-detection of CH_4 and a strong detection of CO in their CRILES+ observations in 2021, with best-fit quenching pressure $\log P_{\text{quench}} = -0.58$. This discrepancy between our detectability trends highlights the potential importance of quenching and other atmospheric processes in shaping the molecular significance of observations. To construct a comprehensive benchmark of detectability across the evolutionary grid, future work should

incorporate a disequilibrium chemistry regime and cloud formation model in the atmospheric models, and explore how these processes modulate the detectability of key molecules across different ages, masses and separations.

6. Conclusions

We have quantified the feasibility of CRIRES+ high-contrast spectroscopy for characterizing young giant planets across a systematic evolutionary grid at 140 pc, spanning 5 and 10 M_{Jup} , projected separations of 24.8–123.9 AU, and ages from $\lesssim 1$ to ~ 65 Myr, using a validated end-to-end pipeline with state-of-the-art likelihood-ratio mapping analysis. Our key findings are:

- Full-spectrum detections above 5σ are achieved at all tested separations for 10 M_{Jup} planets up to ~ 9.5 Myr (typically $\sigma > 10$), declining toward the threshold between ~ 17 and ~ 65 Myr as planets cool and fade. For 5 M_{Jup} planets, simultaneous detections at all separations are limited to $\lesssim 3.65$ Myr, but the widest separation (123.9 AU) sustains detection through ~ 15 Myr, demonstrating that angular separation is a key leverage point for extending the accessible age baseline of lower-mass planets. The noise budget evolves with separation: stellar photon noise dominates at $\lesssim 0.6$ arcsec and directly benefits from AO starlight suppression, while sky background becomes a co-equal noise term at wider separations for older planets ($\gtrsim 6.6$ Myr), setting a fundamental detectability floor independent of AO performance.
- Molecular detectability follows a clear age-ordered sequence. For 10 M_{Jup} planets, CO is accessible at the youngest ages ($\lesssim 5$ Myr), CH₄ opens a detection window from ~ 2 to ~ 10 Myr, and H₂O is the most robust tracer remaining above 5σ at all separations through ~ 17 Myr. For 5 M_{Jup} planets, CO is only marginally detectable at the youngest ages ($\lesssim 1.4$ Myr) and widest separations, while H₂O persists to ~ 15 Myr at 123.9 AU. The age window in which all three molecules are simultaneously detectable, which enabling C/O ratio constraints, spans ~ 2 – 10 Myr for 10 M_{Jup} and narrows to the youngest ages ($\lesssim 1.4$ Myr) for 5 M_{Jup} planets.

These results establish age-resolved detectability benchmarks that map directly onto nearby star-forming regions at ~ 140 pc with different ages. The demonstrated feasibility of molecular detections at angular separations as small as ~ 0.2 arcsec, pushes high-contrast K-band spectroscopy toward a new separation regime previously considered beyond reach with current facilities. As ELT instruments such as HARMONI and METIS expand the directly imaged planet sample into this mass and separation space, the benchmarks and scaling relations presented here provide a practical framework for target selection, observing strategy, and preparatory detectability assessment for the next generation of atmospheric characterization studies.

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Appendix A: Cross-correlation analysis

870 As an independent cross-check of the likelihood-ratio detections described in Section 3, we also implement a cross-correlation analysis. We adopt the cross-correlation method from de Regt et al. 2024 to construct the cross-correlation workflow. First, we compute a residual spectrum R by subtracting the scaled stellar model $\mathbf{m}_s$ from the observed data, then cross-correlate it against molecular templates. We determine the scaling coefficient $[c_s, c_p]$ through non-linear least-squares fitting with `scipy.optimize` (Virtanen et al. 2020). To avoid spurious stellar features in the residual at the planet location, we replace c_s at the off-planet reference position on the other side of stellar PSF. The residual spectrum in order o is thus

$$R_o[x, i] = \mathbf{d}_o[x, i] - c_s \mathbf{m}_{s,o}[x, i], \quad (\text{A.1})$$

where $\mathbf{d}_o$ is the observed data, $\mathbf{m}_{s,o}$ is the stellar model template, x indexes spatial pixels, and i indexes wavelength pixels.

For cross-correlation templates, we use the full model spectra consisting of the three molecules of interest (CO, CH₄ and H₂O) same as the input in ETC to set up a baseline test for detectability. The continuum is removed by a low-pass filter to keep only the line features. To obtain the templates that only contain the contribution from a certain molecule, we generate three other model spectra with the same parameters but with one molecule removed at a time, and then apply the same low-pass filter to get the corresponding continuum-removed templates. Thus:

$$T_X = f_{\text{Full}} - f_{\text{NoX, aligned}} \quad (\text{A.2})$$

for molecule X, where f_{Full} is the full model spectrum and $f_{\text{NoX, aligned}}$ is the model spectrum without molecule X, aligned to f_{Full} to ensure no significant residual continuum offset.

For the dataset, we extract the molecular signal from R by subtracting $f_{\text{NoX, aligned}}$ to get a residual dataset that contains only the contribution from molecule X:

$$R_X = R - f_{\text{NoX, aligned}} \quad (\text{A.3})$$

For each radial velocity v , the processed template is shifted onto the detector wavelength grid as $T_v[i]$. The Pearson correlation coefficient is then

$$r_o[x, v] = \frac{\sum_i R_o[x, i] T_v[i]}{\sqrt{\sum_i R_o[x, i]^2 \sum_i T_v[i]^2}}, \quad (\text{A.4})$$

The order-summed cross-correlation function is

$$\text{CCF}[x, v] = \sum_{o=1}^{N_{\text{ord}}} r_o[x, v], \quad (\text{A.5})$$

and we subtract the median at each velocity to suppress the residual baseline.

890 To characterize template self-similarity and reduce the alias and other correlated noise introduced by the template, we compute the autocorrelation function (ACF),

$$\text{ACF}[\ell] = \sum_i T[i] T[i - \ell], \quad (\text{A.6})$$

which is shifted and normalized to the same peak as the CCF at the planet's position in velocity-separation space. The noise floor σ is then estimated from the residual in the clean velocity region $|v - v_{\text{sys}}| > 100 \text{ km s}^{-1}$ to avoid the wing of CCF. The detection significance is reported directly as

$$\text{SNR}_{\text{Pearson}} = \frac{\text{CCF}[x_p, v_{\text{sys}}]}{\sigma}, \quad (\text{A.7})$$

providing a consistent template-dependent normalization SNR across molecular species.

We adopt the likelihood-converted significance z rather than the CCF S/N as the primary detectability metric throughout this work. The likelihood-ratio test requires no defined signal-free velocity region for noise estimation and directly compares the models, whereas the frequentist interpretation of the CCF relies on assumptions about the noise distribution and the definition of the noise floor, which may be affected by aliases and systematics. We therefore use the cross-correlation analysis only to validate the likelihood results, as demonstrated on the β Pictoris b analog in Appendix B.

Appendix B: Pipeline validation on β Pictoris b

We generated mock CRRES+ observations of a β Pictoris b analog to validate the pipeline, with the system parameters listed in Table B.1. The simulations adopt the same observational setup as Landman et al. (2024): $\text{DIT} \times \text{NDIT} = 15 \text{ s} \times 8$, with 20 frames per nodding position, corresponding to the total integration time of two observing nights, and use the same meteorological parameters as Table 2.

We generate the planetary model spectrum using the best-fit atmospheric retrieval parameters of Landman et al. (2024), listed in Table B.2. The spectrum is computed with pRT-3 (Mollière et al. 2019, 2020), adopting the EddySed cloud model

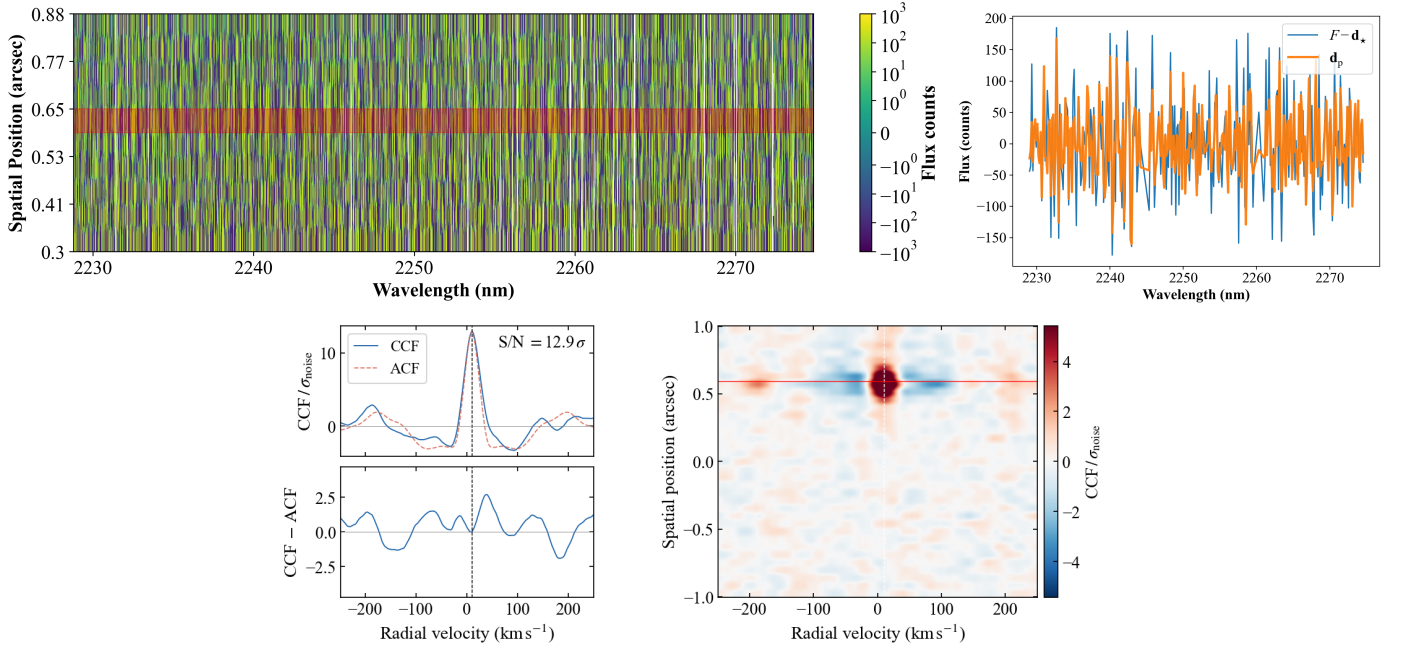


Fig. A.1: Following the same putative system in Figure 5, the left panel shows the 2D spectrum after starlight subtraction, with the planet trace indicated by the red dashed line. The top-right panel plots the extracted 1D residual spectrum (blue) compared with the best-fit planet model (orange) at planet’s position. The bottom-right panel shows the cross-correlation function (CCF) at the planet position.

Table B.1: Adopted parameters for the simulated β Pictoris system. Reference index: (1) Baburaj et al. (2026); (2) Landman et al. (2024); (3) Bonnefoy et al. (2013); (4) Royer et al. (2007).

Parameter	Value	Reference
$T_{\text{eff},\star}$	8000 K	1,3
$\log g_{\star}$	4.0	1,3
$K_{\star}$	3.45	3
K_p	12.7	3
BERV	32 km/s	2
System barycentric velocity	20 km/s	2
$v_p \sin i$	20 km/s	2
$v_{\star} \sin i$	130 km/s	4
Projected separation	0.48 arcsec	2

Table B.2: Parameters adopted to generate the validation-test model spectrum. Values are taken from the best-fit atmospheric retrieval of Landman et al. (2024).

Parameter	Value	Description
[Fe/H]	-0.39	Metallicity
C/O	0.41	Carbon-to-oxygen ratio
$\log(P_{\text{quench}})$	-0.58	Carbon quench pressure (bar)
$\log(g)$	4.23	Surface gravity (cm/s^2)
$\log(X_{\text{Fe}})$	-0.69	Fe cloud condensate mass fraction
$\log(X_{\text{MgSiO}_3})$	-0.67	MgSiO ₃ cloud condensate mass fraction
f_{sed}	5.11	EddySed sedimentation parameter
$\log(K_{\text{zz}})$	9.23	Eddy diffusion coefficient (cm^2/s)
σ_g	2.02	Log-normal particle size distribution width

(Ackerman & Marley 2001) with Fe and MgSiO₃ condensates, and a carbon quenching pressure of $\log(P_{\text{quench}}) = -0.58$ bar that suppresses the formation of CH₄ via vertical mixing (Zahnle & Marley 2014). As a result, the model K-band spectrum is dominated by H₂O and CO absorption, with negligible CH₄ spectral content (Figure B.1). Since Landman et al. (2024) do not provide a direct K-band flux measurement of β Pic b, we adopt the K-band magnitudes for both star and planet from independent photometric studies (Table B.1).

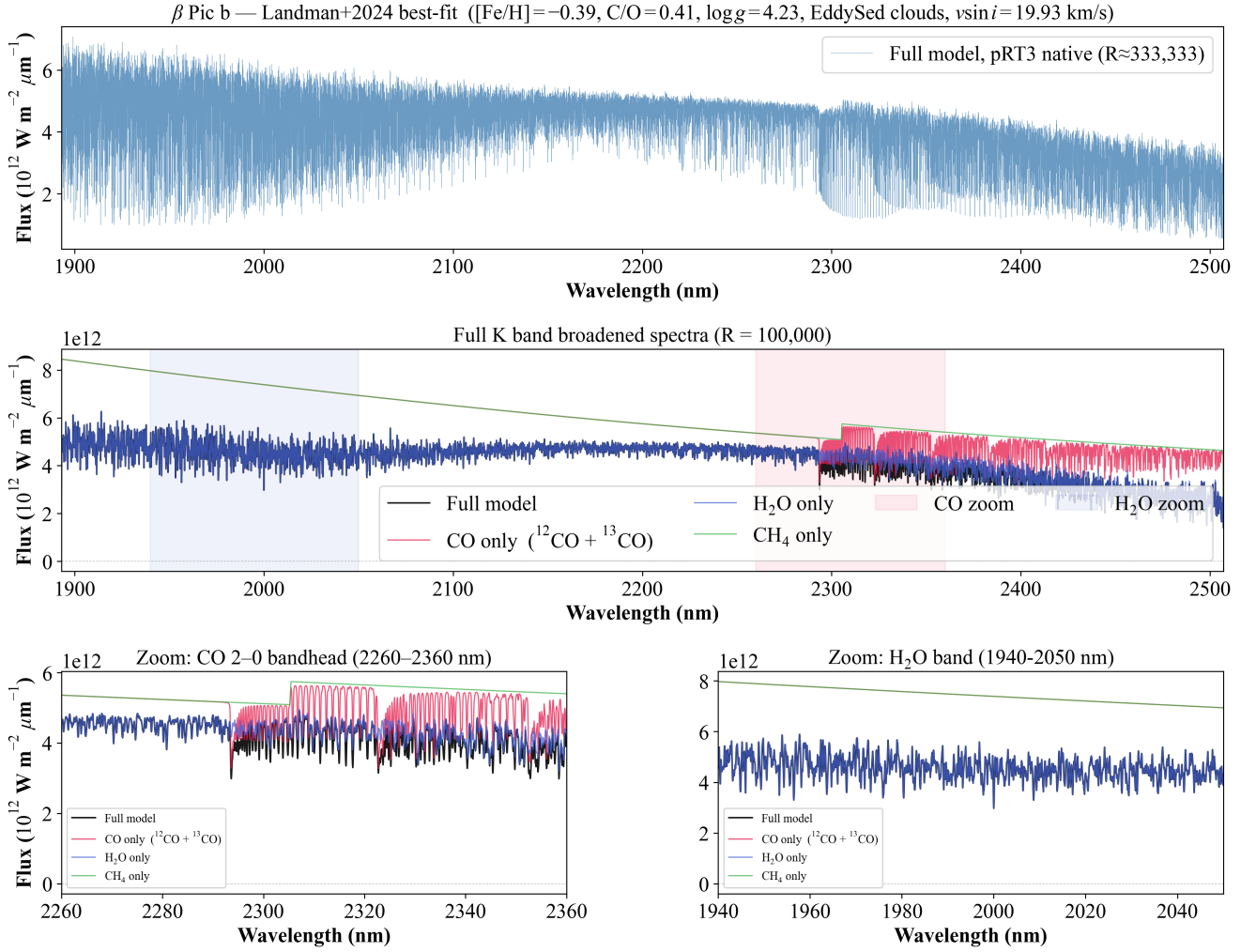


Fig. B.1: High-resolution model K-band spectrum for the β Pictoris b analog. The adopted parameters are listed in Table B.2.

Figure B.2 shows the cross-correlation results for the full-spectrum template and three individual molecular templates. The full-spectrum cross-correlation recovers a clear detection peak with $\text{S/N} \sim 26.2$, consistent with the S/N values reported by Landman et al. (2024) ($\text{S/N} \sim 24.5$). The H_2O template yields a strong detection, likewise consistent with Landman et al. (2024) ($\text{S/N} \sim 24.0$ and 21.0). The CO template produces a cross-correlation S/N of ~ 14.2 , stronger than what reported in Landman et al. (2024) ($\text{S/N} \sim 8.4$). The rightmost panel of Figure B.2 shows the CH_4 cross-correlation, which yields no detection. This is a direct consequence of the injected model — because carbon quenching suppresses CH_4 to negligible abundances in the K band, the model spectrum carries no CH_4 spectral features to cross-correlate against. The non-detection is therefore expected and consistent with the non-detection of CH_4 reported by Landman et al. (2024).

Figure B.3 shows the $\Delta \ln \mathcal{L}$ curve as a function of radial velocity, with a peak exceeding 900 at the injected planet radial velocity. The result is higher than the $\Delta \ln \mathcal{L} \sim 600$ from Landman et al. (2024), which is consistent with the higher S/N obtained in the cross-correlation analysis.

Our simulated detection significances are systematically higher than observations, with $\Delta \ln \mathcal{L}$ overestimated by a factor of ~ 1.6 . This optimism arises from multiple sources: (i) the CRIRES+ ETC overestimates flux counts at the planet position with a factor of ~ 1.8 , possibly due to unmodeled instrumental throughputs and realistic AO efficiency; (ii) our pipeline omits pre-processing steps (flat and bias calibration) that reduce real S/N by introducing additional noise; (iii) we assume constant airmass and seeing, whereas in real observations, telluric absorption and instrumental systematics are time-variable; and (iv) post-processing uses the input model spectrum for cross-correlation, but true planetary spectra may deviate from the model. These effects lead to a discrepancy between S/N of simulated and observed data. Notably, these sources of overestimation may not be universal, and agreement between simulated and observed S/N varies with system properties and observing conditions. Nevertheless, the overall consistency validates the pipeline as a reliable tool for detectability assessment.

Appendix C: Parameters of the evolutionary grid

Parameters selected for the whole evolutionary grid for both $5 M_{\text{Jup}}$ and $10 M_{\text{Jup}}$ planets.

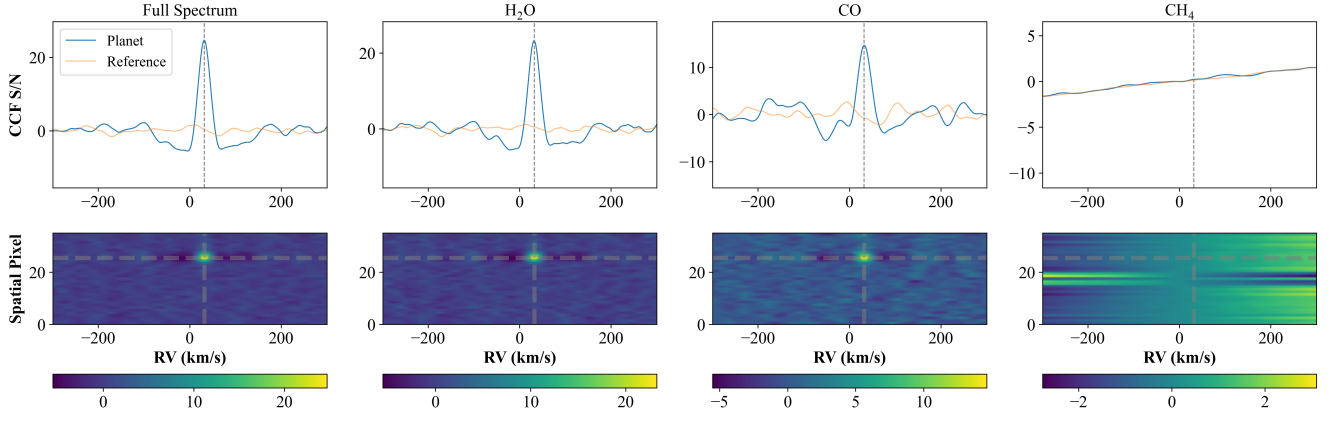


Fig. B.2: Cross-correlation spectra for the β Pictoris b analog. The top row shows the 1D CCF curve converted to S/N at the injected planet position, and the bottom row shows the corresponding 2D CCF map along the spatial axis. From left to right: full-spectrum, H_2O , CO , and CH_4 templates. The CH_4 panel shows no cross-correlation signal, consistent with the carbon-quenched model spectrum in which CH_4 carries negligible spectral power in K band.

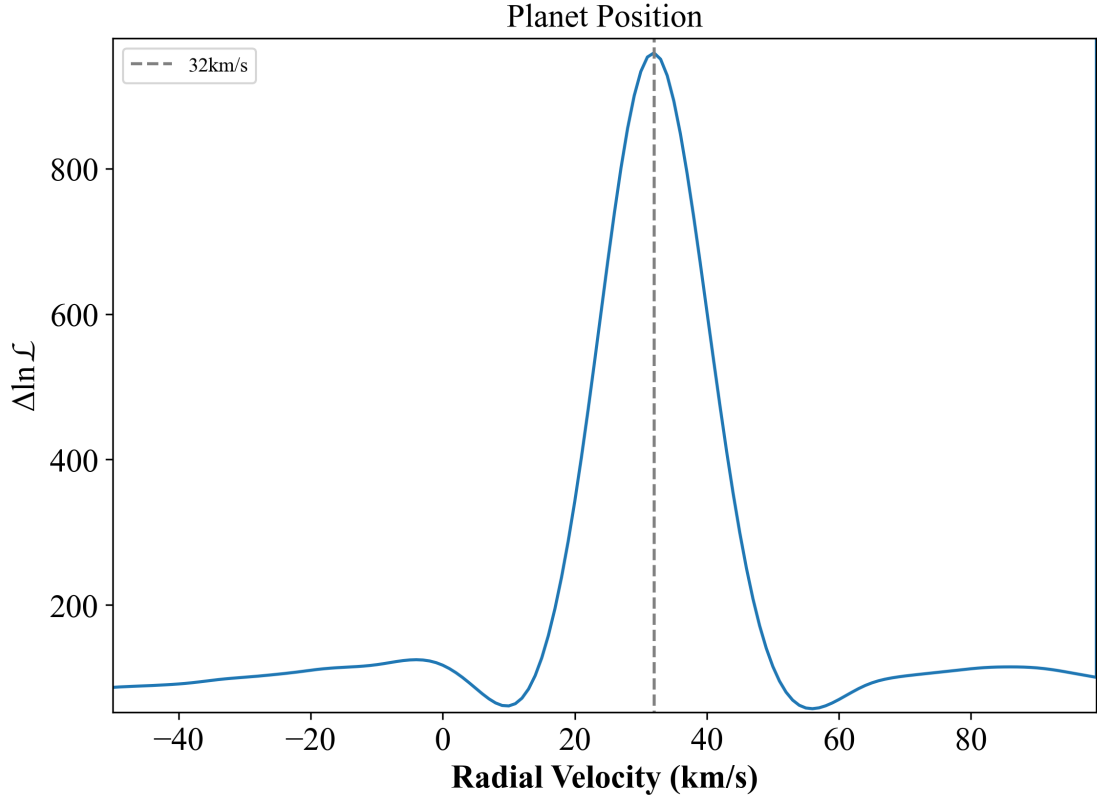


Fig. B.3: The $\Delta \ln \mathcal{L}$ curve as a function of radial velocity for the β Pictoris b analog. The peak exceeds 900 at the injected planet radial velocity.

Table C.1: Planetary system parameters used in this work.

Age (Myr)	M_{planet} (M_J)	$T_{\text{eff,p}}$ (K)	R_{planet} (R_J)	$\log g_{\text{planet}}$ (cgs)	Z_{planet}	K_{planet} (mag)	$R_{\star}$ ($R_{\odot}$)	$Z_{\star}$	$T_{\text{eff},\star}$ (K)	$\log g_{\star}$ (cgs)	RV (km s^{-1})	$K_{\star}$ (mag)
1.37	5	1800	1.76	3.50	0.00	16.64	2.164	0.00	4364	3.767	10.0	7.73
2.20	5	1600	1.63	3.50	0.00	17.55	1.860	0.00	4346	3.898	10.0	8.06
3.65	5	1400	1.53	3.50	0.00	18.43	1.577	0.00	4320	4.042	10.0	8.41
7.44	5	1200	1.42	3.75	0.00	19.26	1.280	0.00	4320	4.223	10.0	8.87
14.9	5	1000	1.33	3.75	0.00	20.05	1.096	0.00	4517	4.358	10.0	9.14
1.00	10	2400	2.30	3.50	0.00	14.41	2.416	0.00	4377	3.671	10.0	7.49
2.48	10	2200	1.84	3.75	0.00	15.27	1.793	0.00	4340	3.931	10.0	8.14
4.86	10	2000	1.59	3.75	0.00	16.16	1.443	0.00	4310	4.119	10.0	8.60
6.60	10	1800	1.51	4.00	0.00	16.98	1.323	0.00	4313	4.194	10.0	8.79
9.51	10	1600	1.44	4.00	0.00	17.82	1.200	0.00	4354	4.279	10.0	9.01
17.40	10	1400	1.35	4.00	0.00	18.71	1.071	0.00	4641	4.378	10.0	9.16
30.15	10	1200	1.27	4.00	0.00	19.51	1.052	0.00	5530	4.393	10.0	8.96
64.58	10	1000	1.21	4.00	0.00	20.26	0.899	0.00	5532	4.530	10.0	9.30